N5

Design and Manufacture
Research pro forma

SQA

Design brief: 2 — secret safe

Name:		SCN:										

Research findings	Page:		Of:	

Notes to candidates

Before you start to write your specification. You must select a brief and read the 'Instructions for candidates' which give details of the skills you must demonstrate and the type of evidence you must provide.

You should speak to your teacher before you start if you are unclear about what is required for this assessment.

You should use information from the brief and findings from the research to complete the specification below.

DESIGN BRIEF 2 — secret safe
The company would like a proposal for a product which allows users to store their valuables safely on a day out.

The product must be inspired by 'summer time', hide the valuables from passers-by and be waterproof.

Specification

The product must:

- allow you to demonstrate appropriate practical skills

FUNCTION

I CARRIED OUT AN INTERVIEW WITH A POTENTIAL END-USER.

WHAT TYPES OF DAYS OUT DO YOU GO ON WHEN ON STAYCATION?

WE LIKE TO VISIT THE BEACH, GO CAMPING AND VISIT THE PARKS.

WHAT VALUABLE ITEMS WOULD YOU LIKE TO BE ABLE TO HIDE WHEN DOING ACTIVITIES ON YOUR DAYS OUT?

ITEMS THAT I WOULD LIKE TO STORE SECURELY INCLUDE MY MOBILE PHONE, MY EAR PODS AND MY WATCH.

YOU SAID "STORE SECURELY". DOES THIS SECRET SAFE HAVE TO BE LOCKABLE?

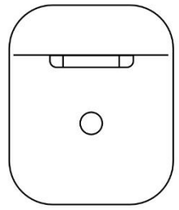
NO, AS OTHER MEMBERS OF MY FAMILY MIGHT NEED TO ACCESS IT, IT'S MORE IMPORTANT THAT THE ITEMS ARE HIDDEN.

WHAT SIZE WOULD YOU LIKE FOR THE SECRET SAFE TO BE?

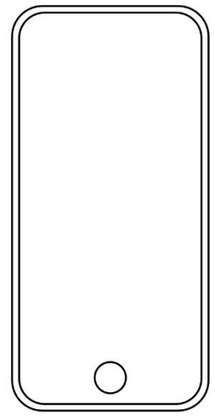
IDEALLY, I'D LIKE TO BE ABLE TO FIT IT INSIDE MY RUCKSACK, ALONG WITH MY OTHER ITEMS.

FUNCTION

MEASUREMENTS OF ITEMS THE PRODUCT SHOULD HOLD.



AIRPODS CASE
Height - 54 mm
Width - 44 mm
Depth - 21 mm



PHONE
Height - 144 mm
Width - 71 mm
Depth - 8 mm



WATCH
Face diameter - 40 mm
Strap diameter - 65 mm
Strap width - 20 mm

DIMENSIONS

MEASUREMENTS OF STANDARD RUCKSACK



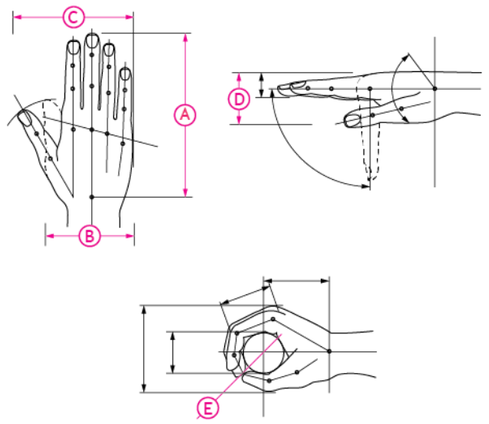
Height - 460 mm
Width - 330 mm
Depth - 210 mm

MATERIALS

I CARRIED OUT A SURVEY TO FIND OUT WHAT THE PRODUCT SHOULD BE MADE FROM.



HAND SIZES



Hand and head		Adult male			Adult female			4-year-old child			16-year-old boy			16-year-old girl		
		Percentile			Percentile			Percentile			Percentile			Percentile		
		5th	50th	95th	5th	50th	95th	5th	50th	95th	5th	50th	95th	5th	50th	95th
A	Hand length	170	190	210	160	175	190	106	116	127	N/A	N/A	N/A	N/A	N/A	N/A
B	Hand breadth	80	90	100	70	80	90	49	55	61	N/A	N/A	N/A	N/A	N/A	N/A
C	Hand span	97	107	116	83	92	101	58	66	73	N/A	N/A	N/A	N/A	N/A	N/A
D	Hand thickness	26	30	34	22	26	30	15	18	21	N/A	N/A	N/A	N/A	N/A	N/A
E	Grip diameter INDEX finger	35	47	55	33	38	44	62	75	87	N/A	N/A	N/A	N/A	N/A	N/A
F	Head height	201	217	232	188	201	214	165	181	198	201	219	237	191	207	223
G	Head length	185	198	210	172	184	196	164	176	188	170	182	193	174	187	200
H	Head breadth	135	144	153	122	131	141	100	109	117	N/A	N/A	N/A	N/A	N/A	N/A